

Urban Pulse - Smart Tourism & Cultural Intelligence Platform

MILESTONE 4 - EQUITY, FORECASTING & FINALIZATION

TEAM - 1

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Problem Statement

Milestone 3 gave us operational visibility into booking behavior, last-mile transport, and weather-driven demand shifts. But three gaps remained:

Transit Equity Gap

Whether heritage sites are equitably accessible by public transport across zones

Fragmented Views

Stakeholders navigating four separate dashboards instead of one unified executive view

Reactive Reporting

Platform reporting on demand after the fact rather than anticipating it

- ❏ **Milestone 4 closes all three:** we measure transit equity, consolidate everything into one executive view, add forecasting and anomaly detection, and finalize the platform for real deployment.

Problem Objectives

1

Transit Equity

Measure how equitably tourists can reach heritage sites via public transit versus other modes

2

Zone Analysis

Identify well-connected versus underserved heritage zones across the city

3

Executive View

Consolidate KPIs across all modules into a single executive-level dashboard

4

Forecasting

Forecast upcoming visitor and transport demand with 7-day projections

5

Anomaly Detection

Detect and flag anomalies such as sudden footfall drops from weather disruptions

6

Deployment

Optimize dashboard performance and prepare the platform for final deployment

Project Scope

In Scope

- Monument Accessibility Index & Transit Coverage Analysis
- Executive Intelligence Dashboard with high-level KPI cards and cross-filters
- Demand forecasting and anomaly detection alerts
- Exportable CSV/PDF summary reports
- Performance optimization and final deployment preparation

Out of Scope

- New Data source integration (uses existing schema tables only)
- Real time live transit tracking
- Features beyond the four defined modules

Proposed Solution

Milestone 4 consolidates our prior milestones into a deployment-ready platform across four modules. *Reuses our existing schema entirely - no new data collection, only new analytical layers.*



Mobility Access Index & Equity Dashboards

Accessibility index using transport data, geospatial equity map, and zone-based comparisons



Executive Intelligence Dashboards

High-level Streamlit overview with top-row KPI cards and interactive cross-filters



Forecasting, Anomaly Detection & Alerts

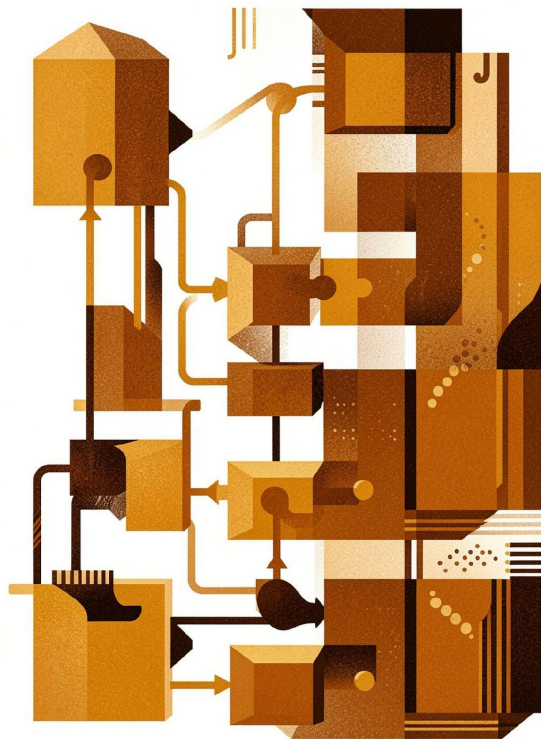
Demand forecasting chart, anomaly logic, and live alert display



Performance Optimization & Reporting Suite

Automated CSV/PDF report generation, caching optimization, and deployment prep

Technology Stack



Streamlit



Executive dashboard framework hosting the consolidated high-level view

Supabase



Existing database queried for equity, forecasting, and anomaly logic

Python



Forecasting and anomaly detection logic built on existing fact tables

CSV/PDF Libraries



Automated report generation for stakeholder downloads

GitHub



Final deployment and version-controlled handoff

About Our Datasets

Milestone 4 introduces no new data sources - it repurposes our existing schema for new analytical layers:

Source	Coverage	Used For
dim_location, dim_weather (existing)	Zone/monument geography, weather patterns	Accessibility Index, Weather Sensitivity Matrix
fact_heritage_transport (existing)	Bus, metro, and other transport trip data	Mobility Access Index, Mode Share Over Time
fact_tourist_bookings (existing)	Historical visitor footfall	Demand Forecasting, Anomaly Detection
All Milestone 1-3 tables (combined)	Full Galaxy schema	Executive Dashboard KPI consolidation

Methodology

Equity Index

Cross-reference transport density with monument zones

Anomaly Detection

Threshold logic flagging footfall deviations



Forecasting

Time-series projection of tourist bookings

Executive Consolidation

Unified KPIs in a Streamlit dashboard

Equity Index

Cross-referencing `fact_heritage_transport` (bus/metro trip density) against `dim_location` (monument zones) to score transit accessibility per site

Forecasting

Time-series projection on `fact_tourist_bookings`, using historical patterns validated in Milestones 2–3 (seasonality, weather sensitivity) to predict 7-day demand

Anomaly Detection

Threshold-based logic flagging footfall deviations that correlate with `dim_weather` events (e.g., heavy rainfall) or festival surges from `fact_festivalparticipation`

Executive Consolidation

All four milestones' KPIs surfaced in one Streamlit interface with shared cross-filters, rather than four disconnected dashboards



Milestone 4 is an integration and intelligence layer - not a new data milestone.

Expected Outcomes



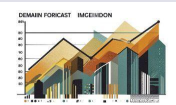
Monument Accessibility Index

Revealing underserved heritage zones across the city



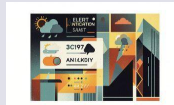
Unified Executive Dashboard

Consolidating KPIs from all four milestones in one view



7-Day Demand Forecasts

With confidence bands, validated against historical patterns



Live Anomaly Alerts

Catching weather-driven or festival-driven demand shifts in real time



Downloadable Reports

CSV/PDF reports for non-technical stakeholders



Deployed Platform

Fully deployed, GitHub-hosted, performance-optimized platform

Conclusion



Milestone 4 transforms Urban Pulse - Smart Tourism & Culture from four separate analytical modules into one coherent, decision-ready platform. By measuring transit equity, consolidating KPIs, forecasting demand, and flagging anomalies, all without collecting a single new dataset, proving our initial schema supported the entire project lifecycle.

- Schema-Driven Equity
Equity analysis built entirely on existing transport and location data
- One Unified View
One unified executive dashboard replacing four separate module views
- Validated Intelligence
Forecasting and anomaly detection grounded in prior validated patterns
- Deployment-Ready
A platform genuinely usable by tourism boards and policymakers

Thank You

Team 1 - Smart Tourism & Cultural Intelligence Platform